

Technical Documentation: Image Processing Mini Projects

1. What I Built:

I built two simple OpenCV-based projects:

- Task 1: Basic Image Manipulation, where I applied grayscale conversion, Gaussian blur, and Canny edge detection on an image.
- Task 2: Color Detector using webcam, which displayed BGR values when clicking on any pixel of the webcam feed.

2. Challenges Faced:

- Initially, I faced difficulties setting up the environment and running Python files from VS Code.
- Understanding how mouse callbacks worked in OpenCV took some time and debugging.
- Some issues occurred due to incorrect file paths or missing files.

3. Improvements with More Time:

- Add functionality to detect the closest named color using a color dataset.
- Add GUI elements using Tkinter or PyQt for a better user interface.
- Save clicked color history and allow exporting it.
- Write modular and well-documented code with comments and better structure.